

The Binomial Theorem

ANSWER KEY



Binomial coefficients

- 1 Evaluate each binomial coefficient:

a $\binom{6}{2} = 15$

b $\binom{8}{0} = 1$

c $\binom{5}{5} = 1$

d $\binom{10}{3} = 120$

- 2 Write down row 4 of Pascal's triangle (the row beginning 1, 4, ...) and explain how it relates to the expansion of $(a + b)^4$.

Row 4 is 1, 4, 6, 4, 1 — these are the coefficients of $a^4, a^3b, a^2b^2, ab^3, b^4$ in the expansion.

Expansions and specific terms

- 3 Expand $(x + 2)^4$ fully.

$$x^4 + 8x^3 + 24x^2 + 32x + 16.$$

- 4 Find the coefficient of x^3 in the expansion of $(2x - 1)^5$.

Term: $\binom{5}{2}(2x)^3(-1)^2 = 10 \cdot 8x^3 = 80x^3$. Coefficient: 80.

- 5 Find the term containing y^4 in the expansion of $(x + y)^7$.

$$\binom{7}{4}x^3y^4 = 35x^3y^4.$$

- 6 Use the binomial theorem to expand $(1 + 0.02)^5$ to the x^2 -style second-order term, and hence estimate 1.02^5 to four decimal places.

$$1 + 5(0.02) + 10(0.02)^2 + \dots = 1 + 0.1 + 0.004 + \dots \approx 1.1041 \text{ (true value } 1.10408\dots).$$